

# Serial Decode — Keithley DMM6500, DAQ6510, DMM7510, SMU2461

A UART decoder that runs **on** a Keithley bench instrument. It digitizes the line with the instrument's own digitizer, recovers the baud rate, frame format and idle polarity from the signal, and shows the bytes on the front panel as text or hex. No host, no logic analyser, one probe.

Ian Ameline · **version 1.21** · MIT licence (see [LICENSE](#))

SERIAL DECODE - 240B FRAME

End App

| BAUD                                            | FORMAT                  | IDLE | LOGIC    | THRESH | SA/BIT | RATE                    | BYTES | ERR              | FIT  | S/N   |
|-------------------------------------------------|-------------------------|------|----------|--------|--------|-------------------------|-------|------------------|------|-------|
| 9600                                            | 8N1                     | HIGH | 3V3 CMOS | 1.63 V | 8.3    | 80 kS/s                 | 239   | 0                | 1.00 | 74 dB |
| Start bit                                       |                         |      |          |        |        |                         |       |                  |      |       |
| 0000                                            | 8B 81 69 6E 20 63 75 6C |      |          |        |        | 70 61 20 71 75 69 20 0D |       | ??in culpa qui . |      |       |
| 0016                                            | 0A 6F 66 66 69 63 69 61 |      |          |        |        | 20 64 65 73 65 72 75 6E |       | .officia deserun |      |       |
| 0032                                            | 74 20 6D 6F 6C 6C 69 74 |      |          |        |        | 20 61 6E 69 6D 20 69 64 |       | t mollit anim id |      |       |
| 0048                                            | 20 65 73 74 20 6C 61 62 |      |          |        |        | 6F 72 75 6D 2E 20 4C 6F |       | est laborum. Lo  |      |       |
| 0064                                            | 72 65 6D 20 69 70 73 75 |      |          |        |        | 6D 20 64 6F 6C 6F 72 20 |       | rem ipsum dolor  |      |       |
| 0080                                            | 73 69 74 20 0D 0A 61 6D |      |          |        |        | 65 74 2C 20 63 6F 6E 73 |       | sit ..amet, cons |      |       |
| 0096                                            | 65 63 74 65 74 75 72 20 |      |          |        |        | 61 64 69 70 69 73 63 69 |       | ectetur adipisci |      |       |
| 0112                                            | 6E 67 20 65 6C 69 74 2C |      |          |        |        | 20 73 65 64 20 64 6F 20 |       | ng elit, sed do  |      |       |
| 0128                                            | 65 69 75 73 6D 6F 64 20 |      |          |        |        | 74 65 6D 70 6F 72 20 69 |       | eiusmod tempor i |      |       |
| 0144                                            | 6E 63 69 64 69 64 75 6E |      |          |        |        | 74 20 0D 0A 75 74 20 6C |       | ncididunt ..ut l |      |       |
| 0160                                            | 61 62 6F 72 65 20 65 74 |      |          |        |        | 20 64 6F 6C 6F 72 65 20 |       | abore et dolore  |      |       |
| 0176                                            | 6D 61 67 6E 61 20 61 6C |      |          |        |        | 69 71 75 61 2E 20 55 74 |       | magna aliqua. Ut |      |       |
| 0192                                            | 20 65 6E 69 6D 20 61 64 |      |          |        |        | 20 6D 69 6E 69 6D 20 76 |       | enim ad minim v  |      |       |
| 0208                                            | 65 6E 69 61 6D 2C 20 71 |      |          |        |        | 75 69 73 20 0D 0A 6E 6F |       | eniam, quis ..no |      |       |
| 0224                                            | 73 74 72 75 64 20 65 78 |      |          |        |        | 65 72 63 69 74 61 74    |       | strud exercitat  |      |       |
| FRAME HEX pg 1/1 bytes 1-239/239 win 240 [done] |                         |      |          |        |        | log: bytes053 719 B     |       |                  |      |       |
| <div>CaptureViewModeNewLogSaveOptions</div>     |                         |      |          |        |        |                         |       |                  |      |       |

The instrument's own front panel, no host attached: 239 bytes of a 240-byte capture at 9600 baud 8N1, no framing errors, S/N 74 dB. The padlock on the BAUD cell means the rate is pinned here rather than re-detected each capture.

|                                   |                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------|
| <a href="#">docs/MANUAL.md</a>    | using it: hooking up, the screen, the buttons, what it copes with, where it fails     |
| <a href="#">docs/REFERENCE.md</a> | every measured number, and the firmware limits behind them                            |
| <a href="#">docs/BENCH.md</a>     | the harness: the seeded sweep, the release gate stage by stage, reproducing a failure |

Ships as `Serial_Decode.tspa`, installed from a USB key through the instrument's own **Manage Apps** screen. Written in TSP — Lua **5.0.2** embedded in the firmware, so the sources avoid `#`, `%`, `string.gmatch` and bitwise operators throughout.

## What to know before trusting it

**Tested on a DMM6500 and nothing else.** Every measured number in these documents came off that one instrument. See [Which instruments](#).

**Flow control is verified electrically but never closed as a loop.** One credit pulse per armed capture, measured on a scope, with nothing waiting for it on the other end. The width is not settable on this firmware, so a receiver needs an edge-triggered input rather than a polling loop. See **Triggering, in both directions** in [REFERENCE.md](#).

**Automatic rate detection has three known failures — 8 points in 860** on a full bench plan for the first two and about 3 in a million for the third — and all three are fixed by typing the rate in: a short pattern repeated over and over can measure at a small multiple of its true rate, a logic low near **−2 V** can misdetect, and above 115 200 baud a long run of one repeated byte can refuse. **None reaches an ordinary payload at a standard baud rate:** across 171 600 captures at standard rates, opened from 200 different points in the waveform, the fox, the 1 kB text payload, twelve random payloads and the walking-bit patterns are clean at every rate from 300 to 250 000 baud. All three are characterised under **Known failures of automatic rate detection** in the [manual](#).

**Nothing stops a long job early.** A touch press cannot be delivered while a script runs, and the front-panel TRIGGER key does **not** deliver `trigger.EVENT_DISPLAY` during a panel-initiated run, so a recording runs to its stated size. That key does still *gate* an armed capture under `Options ▶ Trigger = Trigger key`.

**One press is the whole interaction.** A screenful is ~240 bytes; a recording holds 8 192 or 32 768 bytes to a file, decoded and filed without stepping; beyond that, flow control makes the window a chunk size and credits the device until it stops sending. Ceilings and arithmetic are in the [manual](#).

**It takes over the measurement function and gives it back.** Capturing means digitizing, so every press of Capture selects **Digitize Voltage** on the 10 V range whatever the meter was measuring. The mode is read once at startup and restored when you press **End App**, so ending the app leaves the meter measuring what it was measuring when you launched it, on the range it was on. Two consequences, both deliberate: a function changed from the front panel *while the app is loaded* does not survive the exit — the next Capture reselects the digitizer, so the app keeps working — and the bench soak in `bench/` bypasses this path entirely, so a soak does leave the instrument digitizing.

## Endurance, measured

**Seven and a half days with no computer attached, 136 247 captures, not one instrument event.** 182.2 h, 08-29 to 09-05, twelve segments of the instrument's own bench loop: it read its plan from its USB key, drove the generator over the LAN itself, and wrote every cell back to the key. Every segment's closing total reports **0 events instigated, 0 from the panel** — on this firmware an event is a modal box nothing suppresses — and **0 cells unrecorded in 0 gaps**. The generator needed **270 waveform re-selections**, all recovered: 241 on the second try, 28 on the third, 1 on the fourth.

|                         | duration | cells          | judged BAD | rate          |
|-------------------------|----------|----------------|------------|---------------|
| the week, 08-29 → 09-05 | 182.2 h  | 133 297 judged | 2 462      | <b>1.85 %</b> |
| the night of 09-07      | 17.24 h  | 11 739         | 143        | <b>1.22 %</b> |
| the night of 09-08      | 8.28 h   | 6 708          | 76         | <b>1.13 %</b> |

The last two are the 1 677-point matrix, 7 laps and 4 laps, each on one power cycle; **pooled they give 219 BAD in 18 447 cells, 1.19 %, and they agree with each other within 0.6  $\sigma$ .** The week is **not comparable** with either: it ran an older build *and* a different matrix, laps of 1 326 and 1 333 cells with a different vector set, amplitudes and waits. The drop from 1.85 % is not a measured improvement.

**A low single-digit rate is the expected shape, because the vectors are deliberately hostile:** impulse spikes stacking to 9.3 V on a 3.3 V line, noise and a second signal riding on the logic from generator CH2 summed into CH1 at a sweepable level, baud drifting 6 % and 10 % mid-waveform, 2/10/20 % jitter, an inverted line, LIN break fields no UART frame can legally contain, 256- and 512-byte runs of one value with almost no transitions, payloads longer than the capture — and **every one swept across all 43 rates**, including rates two decades off the one it was built for.

Composition is what stops that being an excuse. Only the seven vectors built to break something may fail — `v47`, `v48a/b`, `j20`, `v61/62/63`, class `loud` in `tools/soakplan.py` — and they carry just **32 % of the week's failures; the other 68 % are exact vectors, where a failure is a defect** and counted as one. On the two current-matrix runs the split is **39 % loud** and **61 % exact** (86 and 132 of 218 pooled). **A confidently wrong byte fails every class**, `loud` included: refusing is a pass, being wrong while claiming to be right never is.

Neither run **leaked anything** — heap after collection at the 17-hour run's six lap seams read 2678, 2678, 2679, 2678, 2679, 2678 and 2678 kB, flat to 1 kB over 10 062 cells, and the 8-hour run's three seams read 2687, 2687, 2687 kB, flat to 0 kB. Lap time did not rise in either, which catches a leak no failure count can. **line is idle** did not occur once in either run, and **generator failures were 0** in both, against 129 in an earlier run — all of those one mis-specified skip list rather than the instrument.

**line is idle** is worth naming because it is real and it is not the app. It is accumulated state in the *instrument*, not the generator, and a soak started shortly after the 13-minute smoke gate can hit it hard — one such attempt went 33 idle in its first 49 cells, with the generator reading back correct throughout. A power cycle of the DMM clears it, and 8.3 hours of running afterwards did not bring it back. It costs a bench run, not a wrong answer: an idle cell is a refusal. See *Instrument hazards* in [BENCH.md](#).

**Confidently wrong bytes, in any of it: none.** Every failure is the app reporting a rate its own framing contradicts, or declining. Separately, over **1 066 400 offline decodes: 95.9 % of failures are the rate being wrong and 0.026 % get a byte wrong with the rate right** — see **Rate detection and decode, counted separately** in [REFERENCE.md](#).

**The offline twin is checked against these runs rather than trusted, and the comparison only means anything at a matched stimulus.** Its default soak deliberately drives half its cells outside the generator's envelope, which a current bench lap never does, so the two must be set side by side on the same plan. Matched, over **444 405 cells in 265 laps** it fails on **1.135 %** against **1.187 %** pooled over 11 bench laps — **0.96x, which is 0.67  $\sigma$  from parity**. Earlier readings of "over-predicting by 1.48x" compared a stressed harness against an unstressed bench and are withdrawn.

**The aggregate matching is partly two errors cancelling**, and the per-vector table in [BENCH.md](#) is what says so: the twin runs four vectors about 1.3x hot and invents ~1.5 failures a lap on random payloads the bench never fails, while scoring zero on the two drift vectors — where it is in fact the *harsher* side, declining 100 % of those cells at the acquisition gate against the bench's 92–98 %. **So it is no longer safely pessimistic**, which is the property this project wants of it, and that is worth knowing before using it as a gate on its own.

## Which instruments

The app needs a digitizer reachable as `dmm.digitize` or `smu.digitize`, the **touchscreen app API** (`display.create` and friends), and **Lua 5.0.2**. Every digitizer below runs at **1 MS/s**, and the **panel is pixel-identical across the whole TSP range**, so the app's screens carry over as they are.

|                        |                                                                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DMM6500</b>         | <b>Tested</b> , firmware 1.7.17a — a 7.6-day run of 136 247 captures with no computer attached and zero instrument events, then 17.24-hour and 8.28-hour runs on the current matrix, and the namespace resolver verified here. 16-bit digitizer, " <i>maximum resolution 16 bits</i> ", specifications, April 2018 |
| <b>DAQ6510</b>         | Should run unmodified, on the strongest grounds of any untested model: it <b>shares the UI board and the acquisition board</b> with the DMM6500, only the channel-board plugin differing. Untested                                                                                                                 |
| <b>DMM7510</b>         | Should run unmodified: 18-bit digitizer, better acquisition boards. Untested                                                                                                                                                                                                                                       |
| <b>SMU2461</b>         | <b>Will install and try; may or may not work.</b> Dual 18-bit digitizers, reached as <code>smu.digitize</code> by a namespace the app resolves at load. That mechanism is verified on the DMM6500; no SMU has ever run it. Three unknowns below                                                                    |
| <b>2470 SMU</b>        | <b>No.</b> " <i>Digitized measurements are not a feature on the 2470</i> " — its reference manual, rev D, October 2024                                                                                                                                                                                             |
| <b>2450 / 2460 SMU</b> | Almost certainly not, by absence rather than denial: neither claims a digitizer, and the 2450 reference (rev D, May 2015) documents <code>smu.measure.*</code> with no digitize function                                                                                                                           |

**Porting is an acquisition question only.** The panel is identical, so the display layer carries over untouched, and the app resolves `dmm.digitize` or `smu.digitize` once at load with no test on any capture path. Every rate figure in REFERENCE descends from the sample rate and the reading buffer's throughput, so the timing figures transfer; resolution is the one axis the app is indifferent to, thresholding each sample into a one or a zero. What to re-measure is what goes *through the acquisition board* — the tolerance envelope, the level and offset limits, the  $-2$  V band. The DAQ6510 shares that board, so a deviation there is a real finding; the DMM7510's differs and those figures may move either way.

**On an SMU2461 the app will install and try, and three things could stop it.** *What the reading buffer holds* — a 2461 digitizes voltage and current **simultaneously** and the decoder trusts consecutive indices, so a buffer carrying current or source values alongside voltage reads as a waveform and surfaces as garbage bytes rather than an error. *The range* — the app pins **10 V**, chosen for the widest digitize bandwidth, which a 2461 may not offer. *The source output* — the app sets it off before every capture and **reads it back, refusing the capture if it cannot confirm it**; those constant names are unverified, and that refusal is the failure it is designed for.

**The `.tspa` header decides where it installs**, independently of whether the code would run: `$Product:` DMM6500, DAQ6510, DMM7510, SMU2461. Neither `DMM7510` nor `SMU2461` appears in the value set Keithley's app-header spec publishes, though Keithley ship TSP apps of their own declaring DMM7510 — an install failure is the symptom if a firmware validates the field strictly, and this field is the first thing to revert.

**If you run this on anything but a DMM6500, please say so** — [open an issue](#). Every row above except the first is inference from API surfaces and vendor documents, and one report from real hardware outweighs all of it. Useful to include: `localnode.model` and `localnode.version`, whether **Manage Apps** offered the app at all, whether both screens built, and whether a capture decoded.

## How to report bugs

**docs/BETA.md** ([PDF](#)) is the note to read first, and the one to hand to anybody trying this on their own bench. It lists the four behaviours that **look** like defects and are not — chiefly that a signal whose band straddles ground is decoded inverted, deliberately — so a tester does not spend their evening rediscovering them.

Then [open an issue](#) with, as far as you can: a **screen grab** of the whole 800x480 frame ( `tools/screenshot.py` takes one over LXI with nothing loaded), the **text that should have been decoded**, the **bit rate** you set, and the **framing** you sent. The expected payload is the one piece that cannot be reconstructed from anything else — it turns "the decode looks wrong" into a difference that can be computed.

A wrong number that looks plausible is worth more than a crash.

## Before releasing it to anyone

Three gates, each about ten times the cost of the one before it. Run them in order.

```
python3 tools/release_sweep.py --offline # ~1 min, no instruments
python3 tools/bench_smoke.py            # 13 min, both instruments
                                         # then POWER CYCLE THE DMM -- see below
python3 tools/soak.py --hours 17 --suites formats,plan --skip-vectors v95,v96,v97
python3 tools/release_sweep.py          # the whole thing, instruments included
```

**Name all three skipped vectors, or the run is not the run you think it is.** `soak.py`'s `--skip-vectors` defaults to empty and the skip is applied *before* the shuffle, so it sets the lap size and keys every amplitude, offset and wait: `v95,v96,v97` gives a **1 677**-cell lap, and naming only `v95,v96` gives **1 720** and lets `v97` back in, where it reads as a generator wedge — that is where an earlier run's 129 "generator failures" came from. `run_bench.py` is the opposite: it already defaults to all three, so do not pass the flag to it at all.

**Power cycle the DMM between the smoke gate and the soak.** A soak started shortly after a smoke can hit the accumulated-state failure described under *Endurance* above; it costs the run, not correctness.

**The smoke gate covers the whole rate range in 13 minutes.** Four waveforms — the fox and a random payload, each in 8N1 and 7E1 — paired crosswise, so one pair takes the 22 standard baud rates and the other the 21 drawn ones, and each rate family faces both formats and both content classes. Then all 45 button presses, seven logic swings from 5 V down to 0.25 V, and eight wrong-locked-rate cases. Measured: 86 cells in 9.7 min, 45 presses in 1.9, 7 levels in 0.7, 8 rate cases in 0.9.

**No version is tagged without at least 8 hours of soak.** A soak reports a **failure rate per test point**, where a sweep reports one pass or one failure. A lap is 1 677 cells and about 2.1–2.5 hours, so 8 hours is the least that separates "fails every lap" from "failed once"; 17 hours gave seven laps and 8.3 gave four.

**Code changes do not get pushed without passing the smoke gate.** On a pass, `bench_smoke.py` writes a receipt holding a hash of `tsp/` and `tools/`; a `pre-push` hook recomputes it and refuses if either tree has moved since. Documentation-only pushes are unaffected — neither tree is hashed — and when the bench is unavailable, `SMOKE_OVERRIDE="reason" git push` proceeds and prints the reason.

```
ln -sf ../../tools/hooks/pre-push .git/hooks/pre-push
```

Every lap draws its waveform order, its non-standard rates and the wait before each capture from the lap number, so `--iteration N` rebuilds any lap exactly without running the ones before it, and a failure replays offline in seconds.

**docs/BENCH.md has the rest:** every stage of the release gate and what it checks, the auditable record a full sweep leaves behind, what state the instruments have to be in before it will start, how to replay a failing cell offline, and the hazards worth knowing.



## Layout

|        |                                                                                                                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tsp/   | the app. <code>serial_core</code> acquisition, <code>uart_decode</code> framing, <code>chunk_decode</code> resumable decode, <code>serial_ui</code> panel, <code>serial_app</code> orchestration |
| bench/ | the soak that runs <b>on</b> the instrument, loaded beside the app for a multi-day run and never shipped inside it. <a href="#">bench/README.md</a> is the runbook                               |
| tools/ | harnesses. <code>release_sweep.py</code> is the entry point; the rest are the authorities it calls                                                                                               |
| docs/  | the manual, instrument references, panel mockups, and <a href="#">BETA.md</a> for testers                                                                                                        |

`tsp/midi_decode.tsp` and `tsp/lin_decode.tsp` are complete and tested but **not shipped** in version 1 — the LIN checksum has never been checked against a real frame. Re-adding either is one line in `tools/package_tspa.py`; the app discovers what it has at runtime.

## Bench

Two instruments over LAN: the DMM6500 under test and an SDG2000X-series generator producing the stimulus. **A scope is optional** — an SDS1000X-E is used here to confirm the stimulus is what it claims to be, and nothing in the gate needs it; it only ever reads. `tools/instruments.py` holds the addresses and the hazards: repeated large waveform uploads wedge the generator's LAN service, and calibration state is off limits on both.

Neither line's top model is needed. The bench asks the generator for 25 MSa/s of TrueArb, 20 Vpp and ~41 stored waveforms, and the scope, if used, for UART decode, two buses and 1 GSa/s; in both lines the model number is the *analog bandwidth*, and the fastest edge here is a 250 kBd bit. An **SDG2042X (40 MHz) and an SDS1104X-E (100 MHz) are sufficient**; the units here are an SDG2122X and an SDS1204X-E. `SDG_MAX_SRATE` is capped at 40 MSa/s so a plan cannot quietly outgrow the cheapest generator; see [docs/BENCH.md](#) for what to check on a unit that is not the one on this bench.